Consultant Profile

Dr. Ayman Younes Allam

Associate Professor & Head of Department – Food Science and Technology

Food Science & Technology Consultant | Functional Foods | Food Preservation | Food Security & Innovation Expert

Shibin El-Kom, Menoufia, Egypt.

+20 102 526 4909 | ayman.younis21@agr.menofia.edu.eg

Google Scholar | Scopus | ORCID | LinkedIn | ResearchGate

1. Academic Profiles:

- **Google Scholar:** https://scholar.google.com/citations?user=_5Xg9jkAAAAJ
- **Scopus Author ID:** https://www.scopus.com/authid/detail.uri?authorId=57414311100
- **ORCID:** https://orcid.org/0000-0002-4439-6508
- **SciProfiles:** https://sciprofiles.com/profile/AymanYouneisAllam
- **LinkedIn:** https://www.linkedin.com/in/dr-ayman-allam-bbb85a97/
- **ResearchGate:** https://www.researchgate.net/profile/Ayman_Allam

1.1 Professional Overview:

I'm a senior academic and applied food science expert with extensive expertise in food technology, functional foods, natural preservation systems, nanotechnology applications, and sustainable food innovation. I integrate advanced scientific research with practical industrial implementation to deliver high-impact, innovation-driven solutions across the food value chain. I have a strong track record in developing and optimizing food systems that enhance product safety, nutritional quality, and shelf-life stability, while reducing post-harvest losses and production inefficiencies. My work focuses on translating cutting-edge research into scalable, industry-ready technologies that directly support national priorities in food security, food safety, and sustainable agricultural development. I provide evidence-based, innovation-oriented consultancy that bridges the gap between academia, industry, and governmental food systems. My expertise contributes to strengthening local food industries, improving export competitiveness, and supporting strategic transformation toward resilient and sustainable food production systems aligned with national development goals.

***Keywords*:** Food Science, Functional Foods, Food Preservation, Nanotechnology, Food Security, Quality Systems, Sustainable Food Processing, Innovation, R&D.

1.2 Strategic Value Proposition:

- Applied food innovation aligned with national food security goals.
- Reduction of food waste and spoilage in perishable commodities.
- Development of clean-label and functional food products.

- Integration of nanotechnology and natural bioactive compounds in food systems.
- Strengthening of food safety and quality assurance frameworks.

2.1 Scientific Expertise:

- Functional food development and nutritional enhancement.
- Food preservation using natural and advanced technologies.
- Nanotechnology applications in food systems (*Nanochitosan, SeNPs*).
- Antioxidant and antimicrobial systems in food matrices.
- Meat and fish product innovation and reformulation.
- Utilization of agricultural and marine by-products.

2.2 Food Safety & Quality Systems

- HACCP system design, implementation, and auditing.
- ISO 22000 food safety management systems.
- Quality control and assurance in food production lines.
- Microbiological and physicochemical evaluation of food products.

2.3 Applied Industrial Skills:

- Shelf-life study design and execution.
- Product development from lab scale to pilot scale.
- Food formulation and process optimization.
- Packaging and preservation system design.

3. Research & Innovation Portfolio:

3.1 Research Focus Areas:

- Natural antioxidants from plant and marine sources
- Nano-encapsulation systems for bioactive compounds
- Clean-label food preservation technologies
- Sustainable food processing and waste valorization

3.2 Key Applied Research Themes:

A. Meat and Fish Preservation:

- Development of natural antimicrobial systems.
- Replacement of synthetic preservatives (*BHT, nitrites*).
- Extension of refrigerated and frozen shelf life.

B. Functional Foods:

- Enrichment of meat products with dietary fibers and bioactives (chia, date pits).
- Development of health-oriented food formulations.

C. Marine and Plant Bioresources:

- Extraction of bioactive compounds from algae.
- Application of plant extracts in food preservation systems.

3.3 Innovation Approach:

- Lab-to-industry translation model.
- Integration of nanotechnology with food systems.
- Focus on scalable and cost-effective solutions.

---

4. Industrial & Governmental Consultancy:

4.1 Government & National Projects Support:

Contributes to:

- National food security programs.
- Agricultural value chain improvement initiatives.
- Food processing and manufacturing development projects.
- Post-harvest loss reduction strategies.

4.2 Industrial Consultancy Services:

- Establishment of food R&D units.
- Optimization of production lines in food factories.
- Development of new food products for commercialization.
- Troubleshooting quality and safety issues in production.

4.3 Training & Capacity Building:

- Training of technical staff in food safety systems.
- Workshops on food preservation technologies.
- Capacity building in functional food development.
- Knowledge transfer for industrial innovation.

5. Impact on Food Security & Sustainability:

5.1 National Food Security Contribution:

- Enhancing availability and stability of food supply.

- Reducing dependence on imported food additives.
- Improving utilization of local agricultural resources.

5.2 Waste Reduction & Resource Efficiency:

- Conversion of agricultural waste into functional ingredients.
- Reduction of spoilage in meat, fish, fruits, and vegetables.
- Improvement of cold chain efficiency through advanced preservation.

5.3 Sustainable Development Alignment:

- Alignment with UN Sustainable Development Goals (SDGs).
- Promotion of eco-friendly food processing technologies.
- Reduction of the environmental impact of food production systems.

6. Selected Applied Achievements:

6.1 Scientific Contributions:

- Development of antioxidant-enriched meat systems.
- Application of algal extracts for microbial control in food products.
- Research in nano-based antimicrobial delivery systems.
- Improvement of shelf-life in meat and fish products using natural systems.

6.2 Practical Applications:

- Functional meat products with improved nutritional profile.
- Natural preservation systems replacing synthetic additives.
- Pilot-scale food formulations with industrial potential.

6.3 Recognition Areas:

- Food innovation and functional product development.
- Nanotechnology applications in food preservation.
- Sustainable food system design.

7. Collaboration Framework:

7.1 Collaboration Interests:

- Governmental authorities (e.g., Future Egypt Authority).
- Food manufacturing industries.
- Agricultural development projects.
- Research institutions and innovation centers.

7.2 Proposed Collaboration Model:

- Joint applied research projects.
- Pilot-scale product development.
- Technology transfer to industry.
- Training and capacity-building programs.

7.3 Vision Statement:

"To transform scientific research into practical, scalable solutions that enhance food quality, safety, sustainability, and national food security through innovation-driven development."

Contact Information:

Department of Food Science and Technology, Faculty of Agriculture, Menoufia University, Shibin El-Kom, Egypt; **+20 102 526 4909** | *ayman.younis21@agr.menofia.edu.eg*